

555 Timer — Monostable Mode

Burglar Alarm Project | Detailed Lesson Guide

■ Lesson Overview	■ Duration	■ Level	■ Kit Cost
555 Timer – Monostable Burglar Alarm	45–60 min	Beginner (Class 6–8)	Under ■50

■ Learning Objectives

By the end of this lesson, students will be able to: (1) Explain the difference between Astable and Monostable modes of the 555 Timer IC. (2) Identify the role of Pins 2, 3, 6, 7, and 8 in Monostable operation. (3) Calculate pulse duration using $T = 1.1 \times R \times C$. (4) Build a simple one-shot burglar alarm circuit on a breadboard. (5) Apply the concept to real-world devices like doorbells and school bells.

■ Materials & Components

Component	Value / Spec	Approx. Cost
555 Timer IC	NE555 or LM555	■5 – ■10
Resistor (R)	10 kΩ (brown-black-orange)	■1
Capacitor (C)	100 μF / 10 V electrolytic	■3 – ■5
Piezo Buzzer	3 V – 5 V	■5 – ■15
Push-button / Trigger wire	Normally Open (NO)	■2
9 V Battery + clip	—	■15 – ■20
Breadboard + jumper wires	Mini or half-size	Shared / reused

Section 1 — Engage: The Burglar Alarm Story

Arjun lives in a small house in Pune. His family wants a simple, affordable alarm to alert them if someone opens the front door at night. They cannot afford an expensive commercial system, but Arjun has just discovered the **555 Timer IC** — a tiny chip that costs less than ₹10 and can do remarkable things!

The key requirement for Arjun's alarm: it should **beep exactly once** when the door is opened, then go silent. This is called a *one-shot* or *monostable* response. Understanding this behaviour is the heart of today's lesson.

■ Discussion Prompt (2 minutes)

Ask the class: Can you think of other devices around you that make a sound just ONCE and then stop? Think about doorbells, ATM beeps, microwave 'done' alerts, or a school bell. What do they all have in common?

Section 2 — Explore: Astable vs Monostable Mode

2.1 Astable Mode — The Non-Stop Drummer

In **Astable mode** the 555 Timer acts like a *dholak* player who never stops beating the drum. The output continuously switches between HIGH and LOW with no external trigger needed. This produces a square wave — perfect for blinking LEDs or generating tones, but not what we want for a single-beep alarm.

Property	Astable Mode
Stable states	0 (always switching)
Needs a trigger?	No — runs automatically
Output	Continuous square wave
Analogy	Ceiling fan — runs until switched off
Typical use	Blinking LED, tone generator, clock

2.2 Monostable Mode — The School Bell

In **Monostable mode** the 555 Timer has one stable state (output LOW / silent). When a trigger signal arrives at **Pin 2**, the output jumps HIGH for a fixed duration, then returns to LOW automatically — exactly like a school bell that rings once and stops.

Property	Monostable Mode
----------	-----------------

Stable states	1 (resting = LOW)
Needs a trigger?	Yes — one pulse at Pin 2
Output	Single timed pulse then LOW
Analogy	School bell — rings once then silent
Typical use	Alarm, timer, debounce circuit

2.3 Side-by-Side Comparison

Feature	Astable	Monostable
Stable state(s)	None	One (LOW)
Trigger required	No	Yes (Pin 2 goes LOW)
Output waveform	Continuous	Single pulse
Duration control	$R_{\square}, R_{\square}, C$	R, C ($T = 1.1 RC$)
Real analogy	Spinning fan	School bell

Section 3 — Explain: Circuit & Components

3.1 The 555 Timer IC — Key Pins

The 555 Timer IC has 8 pins. For monostable operation, four pins are critical. Below is a quick reference:

1 GND Connect to negative terminal of battery (0 V)	2 TRIGGER Hears the burglar! Goes LOW to start the pulse	3 OUTPUT Shouts the alarm — goes HIGH when triggered	4 RESET Usually tied to VCC (Pin 8) to keep it active	5 CTRL Connect 0.01 μ F cap to GND; bypasses noise	6 THRESH Connected to capacitor C — detects when C is charged	7 DISCHARGE Connects to R-C junction; discharges C when done	8 VCC Positive supply voltage (5 V – 15 V; use 9 V here)
---	--	--	---	--	---	--	--

3.2 Component Analogies

Component	Electronics Role	Indian Analogy
Resistor (R)	Controls current flow; limits charge rate of C	<i>Water tap — restricts how fast water flows through</i>
Capacitor (C)	Stores charge; determines how long pulse lasts	<i>Matka (clay pot) — bigger pot = longer fill time = longer alarm</i>
Pin 2 (Trigger)	Starts the one-shot timer when pulled LOW	<i>Pressure mat under the door — burglar steps on it → alarm starts</i>
Pin 3 (Output)	Drives the buzzer HIGH for duration T	<i>Loudspeaker — shouts 'ALARM!' for exactly T seconds</i>
Buzzer	Converts HIGH output into audible sound	<i>The neighbourhood watchman blowing a whistle</i>

Section 4 — The Golden Rule: Timing Formula

The duration for which the alarm buzzes is determined by only two components: the resistor R and the capacitor C. The formula is:

$$T = 1.1 \times R \times C$$

Where: **T** = pulse duration in **seconds** | **R** = resistance in **Ohms (Ω)** | **C** = capacitance in **Farads (F)**

4.1 Worked Example

Worked Example — Standard Alarm

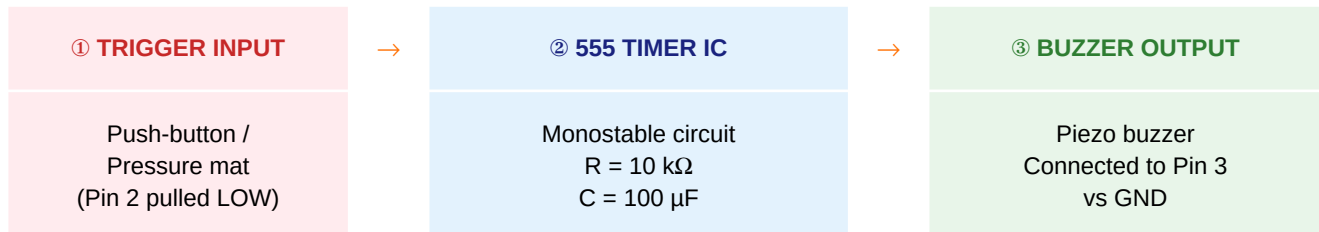
Given: $R = 10,000\ \Omega$ (10 k Ω) | $C = 100\ \mu\text{F} = 0.0001\ \text{F}$ $T = 1.1 \times 10,000 \times 0.0001$ $T = 1.1 \times 1.0$ $T = 1.1$ seconds
The alarm buzzes for approximately 1.1 seconds after the trigger.

4.2 How Changing R and C Affects the Alarm

Change Made	Effect on T	Practical Result
Bigger R (e.g. 100 k Ω)	T increases $\times 10$	Alarm buzzes 10 $\times$ longer
Smaller R (e.g. 1 k Ω)	T decreases $\times 10$	Short quick beep
Bigger C (e.g. 1000 μF)	T increases $\times 10$	Very long alarm duration
Smaller C (e.g. 10 μF)	T decreases $\times 10$	Shorter alarm
Both R and C doubled	T increases $\times 4$	Quadrupled alarm length

Section 5 — Circuit Block Diagram

The burglar alarm circuit can be understood as three blocks working in sequence:



5.1 Step-by-Step Circuit Operation

Step 1. Resting state: Pin 2 is held HIGH (connected to VCC via pull-up resistor). The 555 output (Pin 3) is LOW. Buzzer is silent.

Step 2. Trigger event: A burglar steps on the pressure mat → Pin 2 is pulled momentarily to LOW (below $1/3\text{ VCC}$).

Step 3. Internal flip-flop sets: The 555 internal comparator detects the low voltage at Pin 2 and sets the internal flip-flop, driving Pin 3 HIGH.

Step 4. Capacitor charges: Current flows through R into C. The voltage across C rises exponentially toward VCC.

Step 5. Threshold reached: When C charges to $2/3\text{ VCC}$ (detected at Pin 6), the internal flip-flop resets.

Step 6. Output returns LOW: Pin 3 goes LOW, Pin 7 discharges C to 0 V, and the buzzer falls silent. Circuit is ready for the next trigger.

Section 6 — Real-World Applications

Monostable circuits are everywhere in daily life. Here are some examples your students can relate to:

Application	Trigger Event	Pulse Duration	Component Link
School Bell	Teacher presses button	1 – 5 seconds	<i>$R = 47\text{ k}\Omega$, $C = 100\text{ }\mu\text{F}$</i>
Doorbell Chime	Visitor presses button	2 – 4 seconds	<i>$R = 22\text{ k}\Omega$, $C = 100\text{ }\mu\text{F}$</i>
ATM Keypress Beep	Key is pressed	~0.1 seconds	<i>$R = 1\text{ k}\Omega$, $C = 100\text{ }\mu\text{F}$</i>
Microwave 'Done' Beep	Timer reaches zero	0.5 seconds	<i>$R = 5\text{ k}\Omega$, $C = 100\text{ }\mu\text{F}$</i>
Car Reverse Beep	Gear shifted to reverse	Repeating pulses	<i>Astable drives monostable</i>
Burglar Alarm (ours!)	Pressure mat / door sensor	1.1 seconds	<i>$R = 10\text{ k}\Omega$, $C = 100\text{ }\mu\text{F}$</i>

Section 7 — Quiz Time! ■

Answer the following questions to test your understanding. Answers are provided in the shaded boxes below each question.

Q1. What does the word **MONOSTABLE** mean?

- A. Two stable states
- B. One stable state ✓
- C. No stable state
- D. Multiple states

Answer: B — MONO means ONE. The circuit has one stable state (output LOW). The HIGH state is temporary.

Q2. Which pin on the 555 Timer IC triggers the alarm?

- A. Pin 3
- B. Pin 1
- C. Pin 2 ✓
- D. Pin 8

Answer: C — Pin 2 is the TRIGGER pin. When it is pulled below 1/3 VCC, the one-shot pulse begins.

Q3. What does the capacitor do in this circuit?

- A. Generates sound
- B. Controls timing / stores charge ✓
- C. Amplifies signal
- D. Resets the IC

Answer: B — The capacitor charges through R, and when it reaches 2/3 VCC the pulse ends. Bigger C = longer pulse.

Q4. If we use a **BIGGER** resistor, the alarm will be:

- A. Shorter
- B. Longer ✓
- C. The same
- D. Silent

Answer: B — From $T = 1.1 \times R \times C$, increasing R increases T. Bigger R → longer alarm.

Q5. Calculate the pulse duration for $R = 47\text{ k}\Omega$ and $C = 47\text{ }\mu\text{F}$.

$$T = 1.1 \times 47,000 \times 0.000047$$

Answer: $T = 1.1 \times 47,000 \times 0.000047 = 1.1 \times 2.209 \approx 2.43$ seconds

Section 8 — Fun Homework Assignment ■

Part A — Block Diagram (10 marks)

Draw the block diagram of the monostable burglar alarm from memory, labelling the three blocks (Trigger Input → 555 Timer → Buzzer Output) and at least four pin numbers.

Part B — Find a Monostable Device (10 marks)

Find ONE example of a monostable-like device at home or school. Write two sentences: (1) What is the trigger? (2) For how long does the output stay active? Examples: elevator door-open alarm, fridge-open beep, etc.

Part C — Calculation Challenge (10 marks)

Your teacher wants an alarm that buzzes for exactly 3 seconds. Using $T = 1.1 \times R \times C$ and a fixed capacitor of $C = 100 \mu\text{F}$, calculate the required resistance R . Show your working.

Part D — Creative Extension (Bonus 10 marks)

Sketch or describe how you would modify the circuit so that the alarm can be reset manually (silenced) before T seconds have elapsed. Hint: look up what Pin 4 (RESET) does!

■ Teacher's Answer Key

Part C Answer

$T = 1.1 \times R \times C \rightarrow 3 = 1.1 \times R \times 0.0001 \rightarrow R = 3 / (1.1 \times 0.0001) = 3 / 0.00011 \approx 27,272 \Omega \approx 27 \text{ k}\Omega$ (use nearest standard value: 27 k Ω).

Section 9 — Lesson Recap & Key Takeaways

- **Monostable** means ONE stable state. The circuit rests at LOW and temporarily goes HIGH when triggered.
- The **555 Timer IC** in monostable mode needs only a resistor R and capacitor C to set the pulse duration.
- The **timing formula** is $T = 1.1 \times R \times C$. Bigger R or bigger C → longer alarm.
- **Pin 2 HEARS** the trigger (active LOW). **Pin 3 SHOUTS** the alarm (output HIGH during pulse).
- The capacitor acts like a **matka**: it fills up through R, and when it is 2/3 full (2/3 VCC) the alarm stops.
- Real-world examples include school bells, doorbells, ATM keypads, and microwave timers.
- The full kit costs **under \$50**, making it an affordable hands-on electronics project.

■ Inspiring Thought

Every big invention starts with a small component. The 555 Timer IC, invented in 1972, still powers millions of circuits around the world today — including your burglar alarm! Keep experimenting, keep building.